

# The Agentic AI Bootcamp

## Weekly Build: Claude Plugins & Build Multi-Agent with OpenAI Agents SDK

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### Objective

This week, you will use Claude Cowork with the Data plugin to analyze SaaS data and build a multi-agent system using the OpenAI Agents SDK.

## Part A: AI Data Analyst with Claude Cowork

### Scenario

You are a Data Analyst for a consumer Software as a Service (SaaS) company. The team wants insights from the file "saas\_monthly\_metrics.xlsx". Use Claude Cowork with the "Data plugin" to analyze the dataset and present your findings to leadership.

### Tasks

1. Perform an exploratory analysis and generate a statistical summary that includes the mean, median, and any outliers.
2. Create a single-page dashboard summarizing the dataset.
3. Generate 10 different data visualizations that highlight important trends and patterns.
4. Prepare a 10-slide PowerPoint summarizing your findings for the executive team. The slides should cover the dataset overview, key insights, and key recommendations.

### Deliverables

- Screenshots of your Claude analysis workflow
- The dashboard you created
- The PowerPoint presentation.

## Part B: Creative Advertising Multi-Agent System

### Scenario

1. A marketing team wants to automate campaign ideation using AI agents. Your task is to build a multi-agent system using the OpenAI Agents SDK. Create three agents with different responsibilities: (1) A Creative Director generates 3–5 campaign ideas, (2) A Strategist reviews these ideas, selects the top two, and explains the reasoning, (3) A Copywriter then writes tweets promoting the selected campaigns.
2. Build a pipeline where the output of each agent becomes the input to the next (Creative Director → Strategist → Copywriter).
3. Test the system with the following prompt: "Launch a campaign for a new eco-friendly water bottle in Bali."

### Deliverables

- Submit the completed Jupyter Notebook implementing the multi-agent system.

- Include a one-page explanation describing:
  - The role of each agent
  - Any tools or functions used
  - Why a multi-agent approach improves the workflow

Also include one sample output from the system.

## APPENDIX:

The dataset **saas\_monthly\_metrics.xlsx** contains the following fields:

- **month:** Month and year of the observation (e.g., *Jan-23, Jun-25*).
- **region:** Geographic market where customers are located (e.g., *North America, Europe, APAC, MENA*).
- **plan:** Customer subscription tier (*Basic, Pro, Business, Enterprise*).
- **new\_subscribers:** Number of customers who subscribed during the month.
- **churned\_subscribers:** Number of customers who cancelled their subscription during the month.
- **active\_subscribers:** Total number of active customers at the end of the month.
- **arpu\_usd:** Average Revenue Per User (ARPU) in USD, calculated as total revenue divided by active subscribers.
- **mrr\_usd:** Monthly Recurring Revenue (MRR) generated in that month.
- **net\_revenue\_retention:** Measures revenue retained from existing customers. Values **greater than 1** indicate expansion revenue (e.g., upsells or upgrades).
- **marketing\_spend\_usd:** Monthly spending on marketing activities such as ads, campaigns, and promotions.
- **support\_tickets:** Number of customer support tickets submitted during the month; often a signal of product issues or friction.
- **nps\_score:** Net Promoter Score (NPS), a measure of customer satisfaction and loyalty typically ranging from **0 to 100**.